Sindhura Sri Sarwani Prakki

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.17
FIITJEE Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh)	2020	10.00%
Sikhara EM High School, Mogalrapuram, Krishna District	Class X (Board of Secondary Education, Andhra Pradesh)	2018	10.00%

MASTERS PROJECT

FPGA IP Verification and Hardware-Friendly Block Development for YOLO26:

- Conducted a **literature survey** on **YOLO26** and related detection architectures, building a working understanding of the model's internal computational blocks within the attention mechanism.
- Analyzed the Ultralytics YOLO26 reference implementation to decompose the attention block's dataflow; currently developing a hardware-friendly softmax unit in **Verilog RTL**, with internal blocks verified **bit-exact** against a Python fixed-point reference model.
- Currently working on the design of a generic **FPGA IP verification framework** on the **Xilinx ZC104** board to validate **AXI-Stream** data-transfer and **APB**-based control interfaces across IP blocks.
- Testing primarily involved building **PetaLinux** images tailored to each IP's **XSA/bitstream** output and developing PS-side **C test code** to configure and drive on-board IP verification.

PROJECTS

Design & FPGA Validation of AXI-Stream Quantization/De-Quantization IPs on PYNQ-Z2

- Designed **Verilog RTL** IPs for 16-bit fixed-point to **INT8 symmetric quantization/dequantization**, applying shift-subtract logic and offset rounding to resolve boundary truncation errors.
- Integrated the IPs into a **Zynq SoC** via **AXI-Stream** and an **AXI DMA** controller, enabling continuous streaming with unaligned cross-width transfers.
- Verified hardware performance by streaming 5,000 test samples on-FPGA, achieving a 100% accuracy match confirmed via **Vitis** C drivers and **ILA** traces.

CORDIC-Based Pipelined Division Accelerator (Verilog, AXI4-Lite, Vivado, FPGA)

- Designed and implemented a doubly-pipelined, **CORDIC**-based division accelerator in **Verilog**, using a multi-stage fixed-point architecture for high-throughput arithmetic computation.
- Integrated the accelerator as a custom **AXI4-Lite** IP, building the **Vivado** block design and PS-PL interface with register-based data transfer between processor and FPGA logic.
- Verified functionality through simulation and on-hardware testing on the **ZCU104**, covering fixed-point I/O handling and software control via **Vitis**.

RTL Design & Transistor-Level Implementation of a 64-Bit Custom SPI Converter

- Designed and verified behavioral **Verilog** logic for a 64-bit serial-to-parallel converter, cascading eight 8-bit SPI modules with a hierarchical two-stage D flip-flop architecture for precise serial data shifting.
- Implemented and simulated the design at the **transistor level** using custom CMOS logic in **Cadence Virtuoso**, resolving clock-edge synchronization for accurate parallel data capture.

Design & Simulation of a 4×4 DRAM Array Architecture

- Designed a custom **1T-1C DRAM cell** with a latch-type sense amplifier and pre-charge circuit, demonstrating charge sharing and amplification of millivolt-level differential bitline voltages to full logic swings.
- Developed a 2-to-4 **row decoder** and wordline level shifter to prevent threshold-voltage degradation, improving charge retention and sensing margin.

- Built and verified a complete **4×4 DRAM array** (16 cells) in **Cadence Virtuoso**, validating destructive read, restore, and write operations through transient waveform analysis.

CMOS Two-Stage OTA in gpdk90nm

- Designed and characterized a **two-stage CMOS OTA** in **gpdk90nm** technology using **Cadence Virtuoso**.
- Performed **PVT**, AC, and DC analyses, validating operation across 40°C to 125°C.
- Verified functionality through operating-point analysis, frequency response, and corner-based performance evaluation.

SKILLS

Tools: Xilinx Vivado, Cadence Virtuoso, Synopsys Verdi, LTspice.

Languages: Verilog, C.

Protocols: AXI Stream, AXI Lite.

RELEVANT COURSES

Analog Domain: Analog Electronics, Analog IC Design, Analog IC Design Lab.

Digital Domain: CMOS Digital IC Design, Digital IC Design Lab, Introduction to VLSI Design.

Devices courses: Semiconductor Device modeling, Neuromorphic Devices for AI/ML.

POSITIONS OF RESPONSIBILITY

- Teaching Assistant for the Semiconductors and Device Modeling by **Dr. Kaushik Nayak**
- Teaching Assistant for the Probability and Random Processes courses by **Prof. Anjana AM**

HOBBIES

Sketching, Singing